

Thu Htet

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EDUCATION

University of California, Davis | GPA: 3.38

Davis, CA

Bachelor Of Science – Electrical & Computer Engineering

Expected March 2027

- Relevant courses: Circuits, Embedded Systems, Digital Electronic Circuits, Testing and Verification of Digital Systems, Digital Systems, Computer Architecture, Digital Signal Processing

EXPERIENCE

Space and Satellite Systems — Battery Management System

December 2024 - November 2025

- Designed a 4S lithium-ion battery management system with battery protection, cell monitoring, balancing, charging, and power distribution for CubeSat applications
- Created schematics and PCB layouts in KiCad, integrating protection ICs, sensing circuitry, and power distribution networks while following PCB design best practices
- Built and validated hardware prototypes through soldering, board bring-up, electrical measurements, debugging, and environmental testing, including thermal, short-circuit, and vibration evaluations

Assista Robotics — Undergraduate Robotics Researcher

January 2025 - July 2025

- Developed embedded firmware for a 9-axis robotic arm platform, implementing control logic and system integration for autonomous fryer and grill automation in commercial kitchen environments
- Integrated and debugged embedded software with sensors, actuators, and robotic subsystems, refining system performance through iterative hardware and firmware testing
- Presented the robotic platform at the 2025 UC Davis ECE Picnic Day Showcase, earning 1st Place

De Anza Computer Science Department — Teaching Assistant

January 2023 - June 2024

- Assisted over 100 students across Java, C++, Data Structures, Algorithms, and Object Oriented Programming courses during labs and office hours
- Debugged programming assignments, explained software engineering concepts, and helped students improve problem-solving and coding practices
- Worked closely with instructors to support grading, answer technical questions, and create an effective learning environment

PROJECTS

Smart Home Automation System - PCB Design & Embedded Systems

June 2026

- Designed a modular smart home control interface featuring buttons, rotary encoders, and magnetic expansion modules for configurable device control
- Developed a compact PCB integrating a microcontroller, user input components, and external interfaces to support flexible expansion with additional modules
- Implemented and debugged I2C communication between the microcontroller and peripheral modules, validating hardware and firmware integration during system testing.
- Received UC Davis ECE Outstanding Senior Design Project Award

Smart Fridge Device — Embedded Firmware & Circuit Testing

February 2026

- Developed embedded C firmware on a PSoC microcontroller to monitor refrigerator door status and temperature, generating real-time visual and audio alerts
- Integrated thermal and magnetic sensors with ADC and GPIO peripherals to enable real-time environmental monitoring and event detection
- Configured and debugged I2C, GPIO, timers, and ADC peripherals in PSoC Creator, validating hardware–software interaction on prototype hardware

SKILLS

- **Hardware Design & Test:** KiCad, Altium, Cadence Virtuoso, LTspice, PCB Layout, Schematic Design, Soldering, Oscilloscope, Multimeter, Logic Analyzer
- **Programming/Embedded:** C, C++, Python, Java, Embedded C, Verilog, ESP32, STM32, PSoC, FreeRTOS, I2C, SPI, UART, GPIO, ADC, Timers, Interrupts